Examiner only

4. In 2015, a householder paid £780 for electricity and £950 for gas. On 1 January 2016 solar panels were installed on the roof to provide **hot water** for the house. The cost of the solar panels was £4 680. In 2016, the householder paid £830 for electricity and £840 for gas.

(a) (i) Gas was used to heat the water in the house in 2015. State how the figures allow you to come to this conclusion. [1]

....

....

(ii) Calculate the expected payback time to recover the cost of the solar panels. [2]

Payback time = ... years

(iii) Explain how an increase in the cost of a unit of electricity **and** a unit of gas would affect the payback time. [3]

....

....

....

....

....

(b) During the two years, the cost of a unit of electricity did not change but the number of **extra** units **of electricity** used in 2016 was 300 kWh. Use an equation from page 2 to calculate the cost of a unit of electricity (in pence) during 2015-2016. [3]

Cost of unit = ... p

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